

run_predictive_ablation_cv.R

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run_predictive_ablation_cv
Run Predictive Ablation Cross-Validation

Description

Runs full, no-abiotic, no-spatial, and no-associations sjSDM cross-validation variants and returns fold-level predictive metrics.

Usage

```
run_predictive_ablation_cv(  
  data_model_input,  
  cv_indices,  
  config_model_fitting = NULL,  
  iter = NULL,  
  sampling = NULL,  
  step_size = NULL,  
  device = "gpu",  
  n_cores = NULL,  
  tune = "grid",  
  tune_steps = 20L,  
  selection_metric = "loss",  
  alpha_cov = 0.5,  
  alpha_coef = 0.5,  
  alpha_spatial = 0.5,  
  lambda_cov = 0,  
  lambda_coef = 0,  
  lambda_spatial = 0,  
  cv_fn = sjSDM::sjSDM_cv,  
  seed = 900723L,  
  verbose = TRUE  
)
```

Arguments

<code>data_model_input</code>	Model input list with <code>'data_community_to_fit'</code> , <code>'data_abiotic_to_fit'</code> , and <code>'data_spatial_to_fit'</code> .
<code>cv_indices</code>	Repeated fold lists from <code>'make_repeated_cv_indices()'</code> .
<code>config_model_fitting</code>	Optional fitting configuration list. Used for <code>'error_family'</code> , <code>'use_age_in_formula'</code> , <code>'n_iter'</code> , <code>'n_sampling'</code> , and <code>'n_step_size'</code> .
<code>iter</code>	Optional epoch count overriding <code>'config_model_fitting\$n_iter'</code> .
<code>sampling</code>	Optional Monte Carlo sampling count overriding <code>'config_model_fitting\$n_sampling'</code> .
<code>step_size</code>	Optional batch size overriding <code>'config_model_fitting\$n_step_size'</code> .
<code>device</code>	Device passed to <code>'sjSDM::sjSDM_cv()'</code> . Default is <code>"gpu"</code> .
<code>n_cores</code>	Optional number of cores for <code>'sjSDM::sjSDM_cv()'</code> . Default <code>'NULL'</code> keeps the smoke run sequential.
<code>tune</code>	Tuning strategy passed to <code>'sjSDM::sjSDM_cv()'</code> . Default is <code>"grid"</code> .
<code>tune_steps</code>	Number of random tuning steps when <code>'tune = "random"'</code> .
<code>selection_metric</code>	Single character string. Metric used to select one tuning row per fold and variant. One of <code>"loss"</code> , <code>"auc_test"</code> , or <code>"auc_macro_test"</code> .
<code>alpha_cov, alpha_coef, alpha_spatial</code>	Alpha tuning values passed to <code>'sjSDM::sjSDM_cv()'</code> .
<code>lambda_cov, lambda_coef, lambda_spatial</code>	Lambda tuning values passed to <code>'sjSDM::sjSDM_cv()'</code> .
<code>cv_fn</code>	Cross-validation function. Defaults to <code>'sjSDM::sjSDM_cv()'</code> and is injectable for unit tests.
<code>seed</code>	Single integer seed passed to sjSDM fitting.
<code>verbose</code>	Logical. If <code>'TRUE'</code> , progress messages are printed via <code>'cli'</code> .

Value

A tibble with fold-level metrics for every repeat and variant.

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